LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	28
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Longitudinal dynamic stability, lateral dynamic stability
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Longitudinal dynamic stability, lateral dynamic stability.
5. Specific Objectives	6.1.4.1 Trainee will be able to describe aircraft stability in all the three axis.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of static stability concepts; Instructor supervision required for theoretical verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define longitudinal static stability.
(b) List the three axes of aircraft motion.
(c) Explain the role of the horizontal stabilizer in pitch.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Longitudinal Dynamic Stability • Phugoid Oscillation characteristics • Short Period Oscillation characteristics • Damping effects in pitch motion • Stability criteria for longitudinal modes	L & Ds	PPT, IB	15 min
(b)	• Definition of Lateral Dynamic Stability • Dutch Roll Oscillation characteristics • Roll subsidence characteristics • Spiral mode stability analysis • Coupling effects in lateral motion	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Static vs Dynamic Stability • Importance of damping in flight control • Summary of stability axes • Review of oscillation modes • Preparation for directional stability	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Differentiate between phugoid and short period oscillations.
(b) Identify the primary oscillation mode associated with lateral dynamic stability.
(c) Explain why damping is critical for dynamic stability.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of longitudinal and lateral dynamic stability discussed in this lesson will assist in understanding directional static stability and propeller effects covered in LP 29.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Directional static stability, prop effects, damping in Yaw, directional control, Effect of wing sweep back, spin, lateral dynamic stability.	Refer to the master syllabus reference list	LP 29

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min